26900412 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSEPHY303 - LASER, NON-LINEAR OPTICS AND FIBER OPTICS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

1. State any two key operating characteristics of a ruby laser. [U] / [CO1]
2. Explain Stimulated emission process in a laser system. [U] / [CO1]
3. State how acceptor doping alters the position of the Fermi level in a semiconductor. [U] / [CO1]
4. Under what condition does resonance occur in an open resonator? [U] / [CO1]
5. Explain the working principle of a mechanical Q-switch in a laser cavity. [U] / [CO2]
6. What is a laser amplifier? [K] / [CO2]
7. State why second-harmonic generation (SHG) cannot occur in centrosymmetric media. [U] / [CO3]
8. Write down the frequency relation governing the generation of an idler wave in parametric processes and define each term. [U] / [CO3]

9. What is the purpose of an erbium-doped fibre amplifier (EDFA) in long-distance optical fibre communication? [U] / [CO4]

10. How is photosensitivity at ~240nm utilized to fabricate periodic gratings in germanium-doped silica fibres? [U] / [CO4]

11. Define the acceptance angle of an optical fibre. What is meant by the acceptance cone? [U] / [CO4]

12. What is meant by air-guiding type photonic crystal fibre? [K] / [CO4]

[2x10 = 20]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

13. For a plane electromagnetic wave propagating in a rectangular cavity, show that the wave vector ***k*** is perpendicular to the electric field ***E(t)***. [U] / [CO1]

14. Describe the construction, working principle, and key operational characteristics of a He-Ne laser with a neat energy-level diagram. [U] / [CO1]

15. Explain the working of a semiconductor laser with a suitable energy level diagram. [U] / [CO1]

16. Discuss the principle and operational mechanism of a Kerr-cell Q-switch in a laser system. [U] / [CO2]

17. Explain the phase-matching condition in second-harmonic generation and show mathematically how it leads to the equality of refractive indices at both frequencies ($\eta_{\omega} = \eta_{2\omega}$). [U] / [CO3]

18. Explain the physical mechanism of self-focusing in a nonlinear medium using the Gaussian intensity profile of a laser beam. [U] / [CO3]

19. A step-index optical fibre has core radius a = 25μm, $n_1 = 1.520$, $n_2 = 1.505$, and $\lambda_0 = 2\mu m$. Calculate the V-number and determine whether the fibre is single-mode or multimode. Estimate the number of propagating modes. [A] / [CO4]

20. Analyse how the boundary conditions of a slab waveguide restrict the possible electromagnetic field patterns and lead to the formation of discrete modes. [An] / [CO4]

[6x5 = 30]